

Methods Appendix:

Estimating Local LGBTQ Populations

March 1, 2026

1 Measurement in HPS Microdata

Let $Y_i \in \{0, 1\}$ indicate whether individual i identifies as LGBTQ. Sex is defined using sex assigned at birth, and age is inferred from year of birth and top-coded at 62.

An individual is classified as LGBTQ if sexual orientation is not straight or if gender identity is transgender. Subcategories include lesbian or gay, bisexual, queer, and transgender.

Survey weights (w_i) are retained throughout. At this stage, estimates are purely survey-based and reflect sampling variability.

2 Soft Calibration to Gallup Marginals

Household Pulse Survey (HPS) and Gallup estimates differ in survey mode, sampling variation, and field timing. Rather than imposing exact agreement, HPS-based probabilities are softly calibrated toward Gallup benchmarks.

Each individual begins with a baseline probability:

$$p_{0i} = \text{clipped}(Y_i).$$

A logit adjustment model is estimated:

$$\text{logit}(p_i) = \text{logit}(p_{0i}) + X_i\theta,$$

where X_i includes generation effects and generation-by-sex interactions.

The parameter vector θ minimizes a weighted objective combining: (i) a deviation penalty that keeps estimates close to HPS baselines, and (ii) a margin penalty that discourages deviation from Gallup targets. Optimization is performed using BFGS.

3 State-Cohort LGBT Rate Estimation

Using calibrated probabilities, state-sex-age LGBT identification rates are computed as:

$$\hat{P}(Y = 1 \mid S, \text{State}, A) = \frac{\sum_i w_i p_i^{\text{adj}}}{\sum_i w_i}.$$

Effective sample size is defined as:

$$n_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

Weekly survey waves are averaged to reduce short-term noise.

4 Age-Local Dirichlet Smoothing of Gender Composition

This step estimates the distribution of gender identity (cisgender, non-binary, transgender) within each sexual orientation category, sex assigned at birth, and exact age.

Let $g \in \{\text{cis}, \text{non-binary}, \text{trans}\}$ index gender categories. Define:

$$\hat{p}_{g,\ell,s,a} = \frac{n_{g,\ell,s,a}}{\sum_g n_{g,\ell,s,a}},$$

where $n_{g,\ell,s,a}$ denotes survey-weighted counts.

Effective sample size is:

$$n_{\text{eff},\ell,s,a} = \frac{(\sum w_i)^2}{\sum w_i^2}.$$

4.1 Age-Local Prior Construction

For a focal age a , neighboring ages $a' \neq a$ receive weights:

$$w_{a'}^{(a)} = n_{\text{eff},\ell,s,a'} \exp\left(-\frac{|a - a'|}{\tau}\right),$$

where $\tau > 0$ controls smoothness.

The age-local prior mean is:

$$m_g^{(a)} = \frac{\sum_{a' \neq a} w_{a'}^{(a)} \hat{p}_{g,\ell,s,a'}}{\sum_{a' \neq a} w_{a'}^{(a)}}.$$

4.2 Dirichlet Moment Matching

A Dirichlet prior is centered at $\mathbf{m}^{(a)}$. The concentration parameter is estimated via method of moments:

$$\alpha_0 = \text{mean}_g \left(\frac{m_g(1 - m_g)}{v_g} - 1 \right),$$

where v_g denotes weighted variance across neighboring ages. The prior vector is $\alpha_g = \alpha_0 m_g$.

4.3 Posterior Shrinkage

The smoothed estimate for focal age a is:

$$\tilde{p}_{g,\ell,s,a} = \frac{n_{\text{eff},\ell,s,a} \hat{p}_{g,\ell,s,a} + \alpha_g}{n_{\text{eff},\ell,s,a} + \alpha_0}.$$

5 Cross-State Shrinkage of LGBT Subcategory Shares

For each sex s , age bin a , and LGBT category ℓ , define:

$$\hat{\pi}_{\ell,s,a,j} = P(\ell \mid Y = 1, s, a, \text{state } j),$$

with effective sample size $n_{\text{eff},j}$.

A beta prior is estimated across states using method of moments:

$$\pi_{\ell,s,a,j} \sim \text{Beta}(\alpha, \beta).$$

Posterior shrinkage yields:

$$\tilde{\pi}_{\ell,s,a,j} = \frac{n_{\text{eff},j} \hat{\pi}_{\ell,s,a,j} + \alpha}{n_{\text{eff},j} + \alpha + \beta}.$$

Final shares are renormalized within each state–sex–age bin.